

Heart Disease Analytics — Executive Overview

920

Patients

four clinical sources

55.3%

Disease prevalence

sample estimate

54

Median age

years

78.9%

Male share

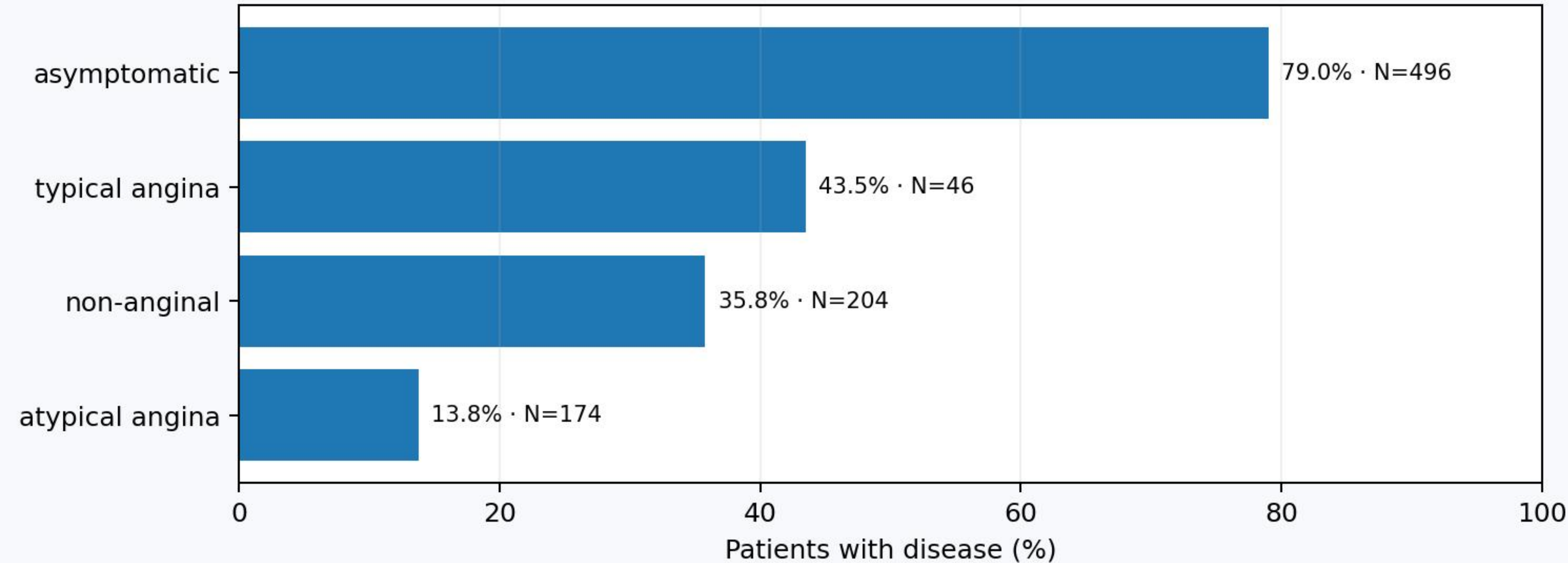
sample imbalance

0.892

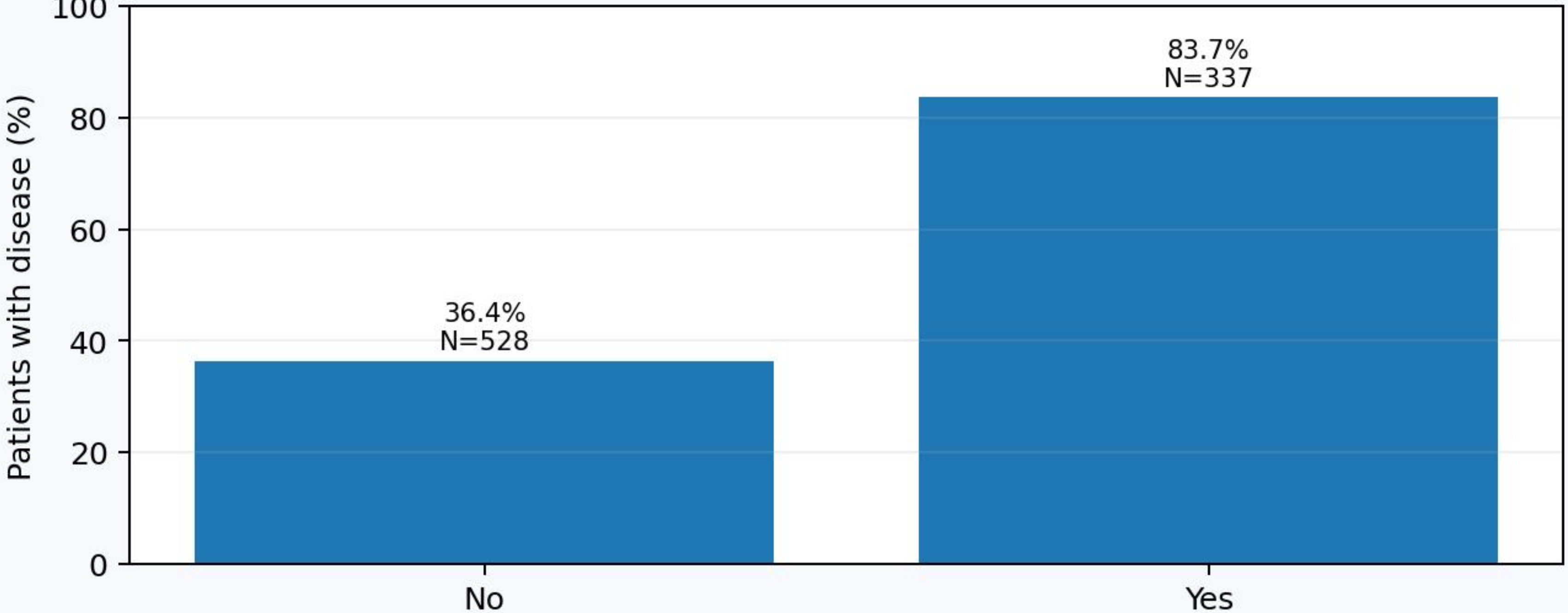
ROC-AUC

internal holdout

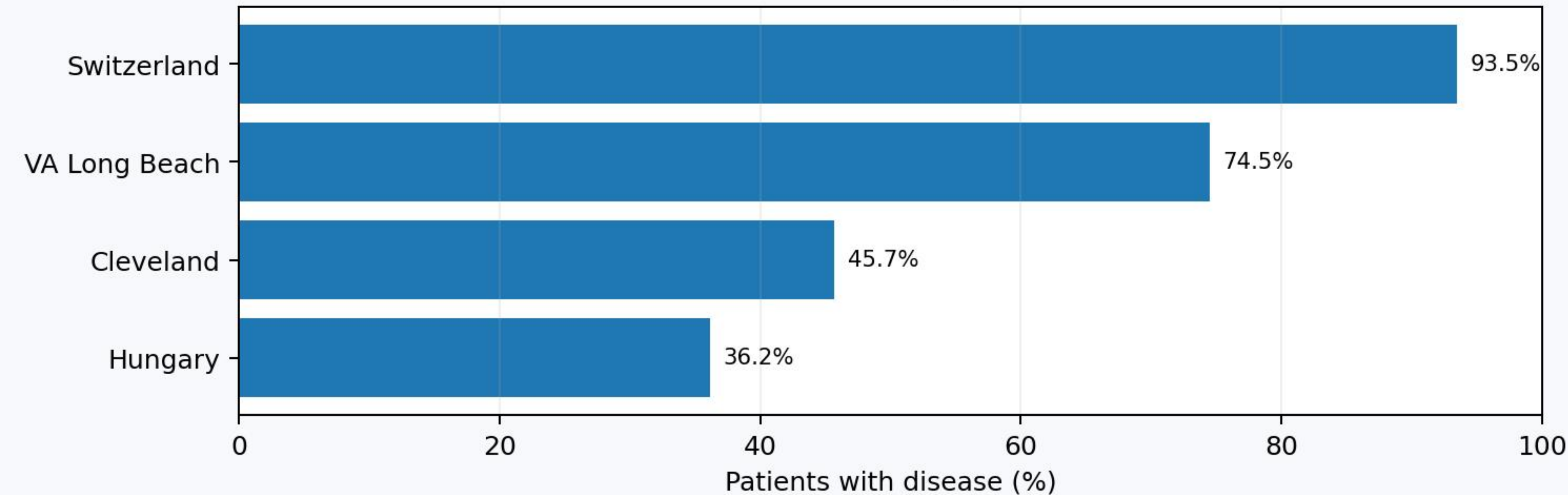
Disease prevalence by chest pain type



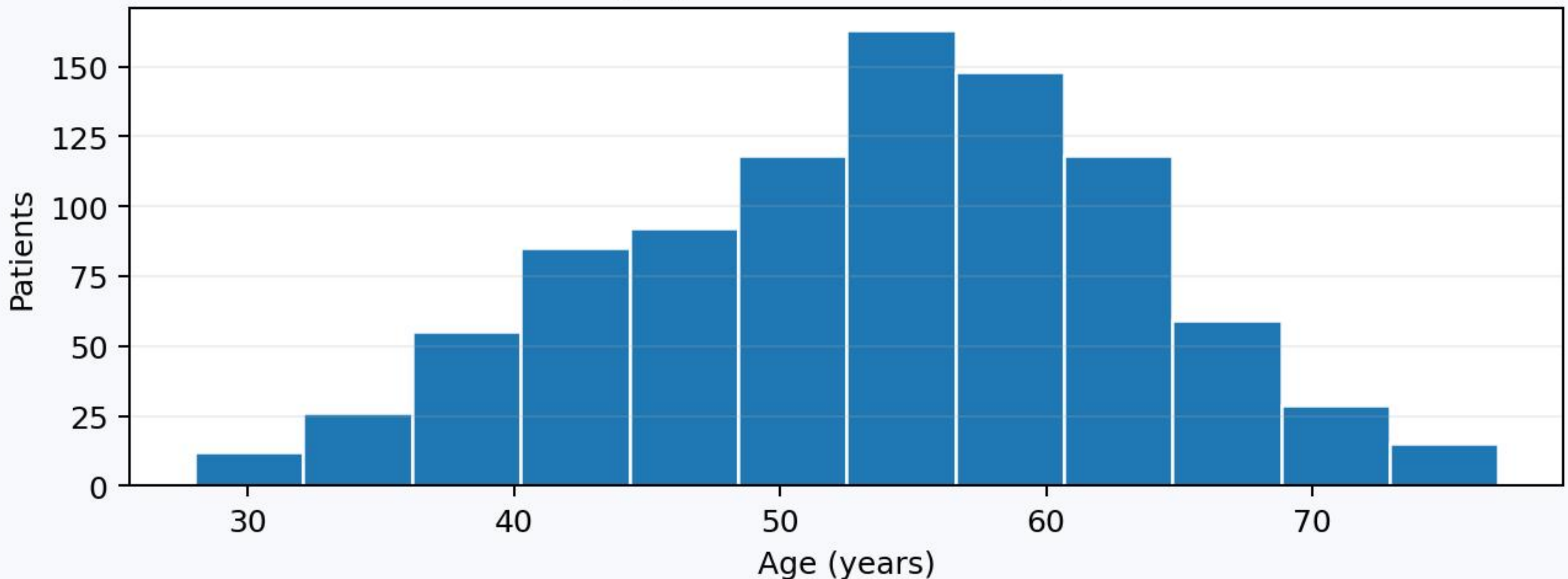
Disease prevalence by exercise-induced angina



Prevalence varies by data source



Age profile of the sample

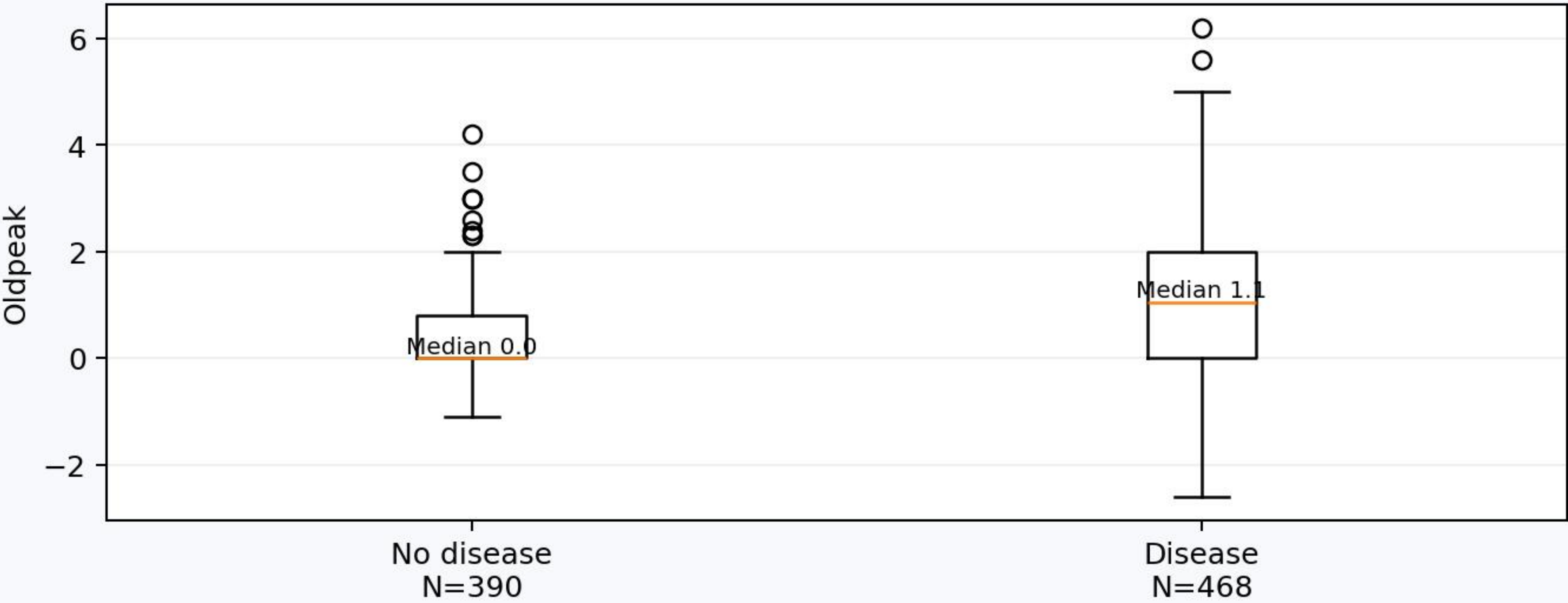


Decision use: identify the strongest descriptive patterns, then verify them by source before acting.

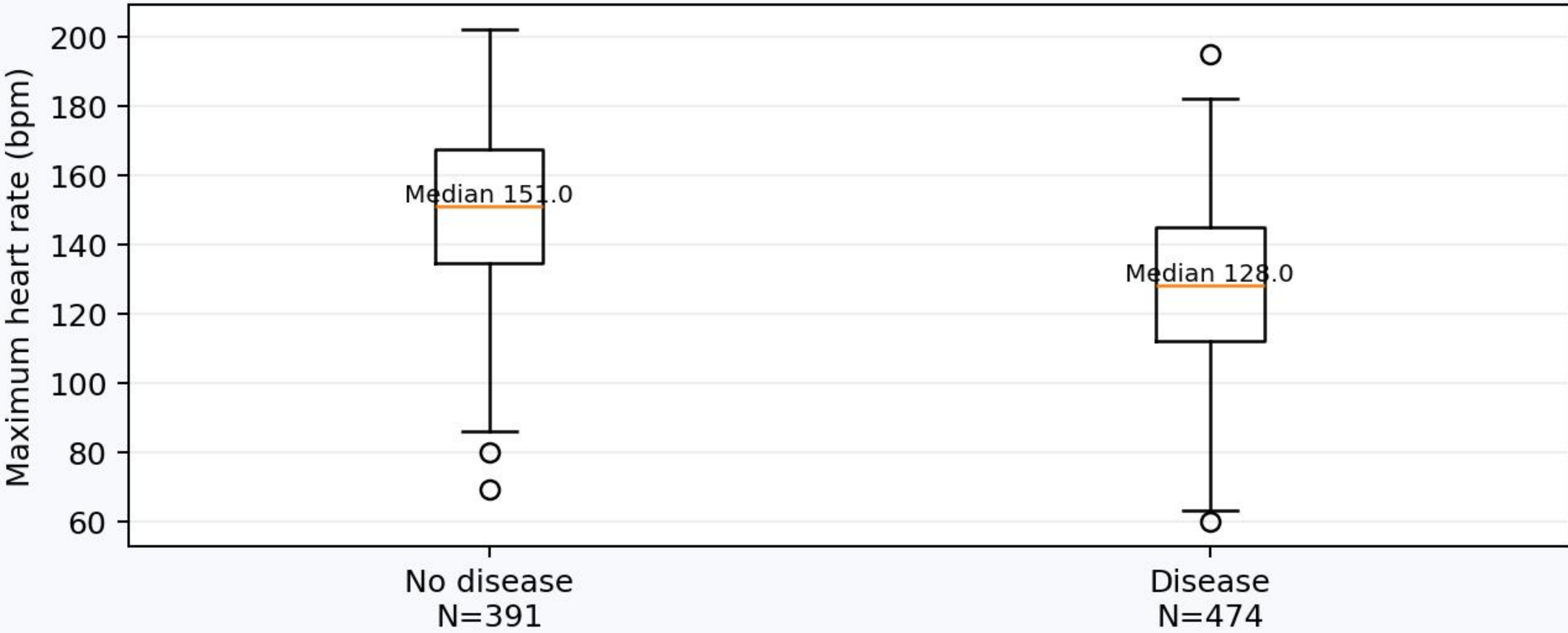
Heart Disease Analytics – Clinical Insights



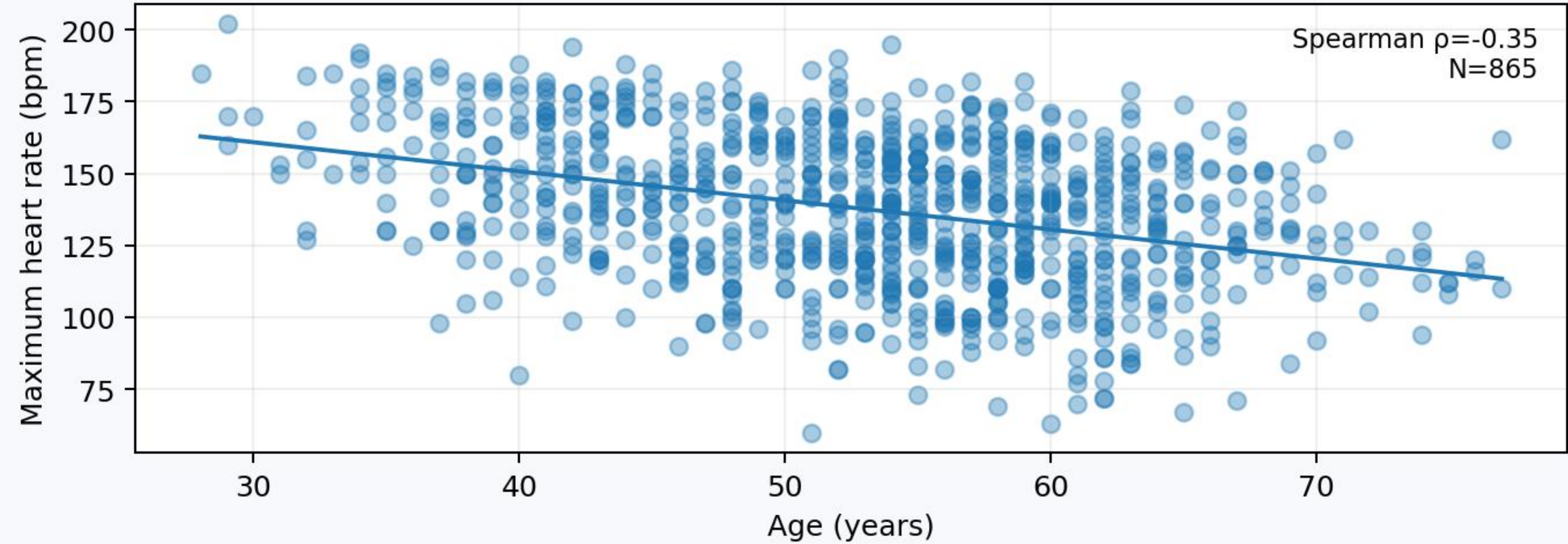
Oldpeak differs meaningfully by disease status



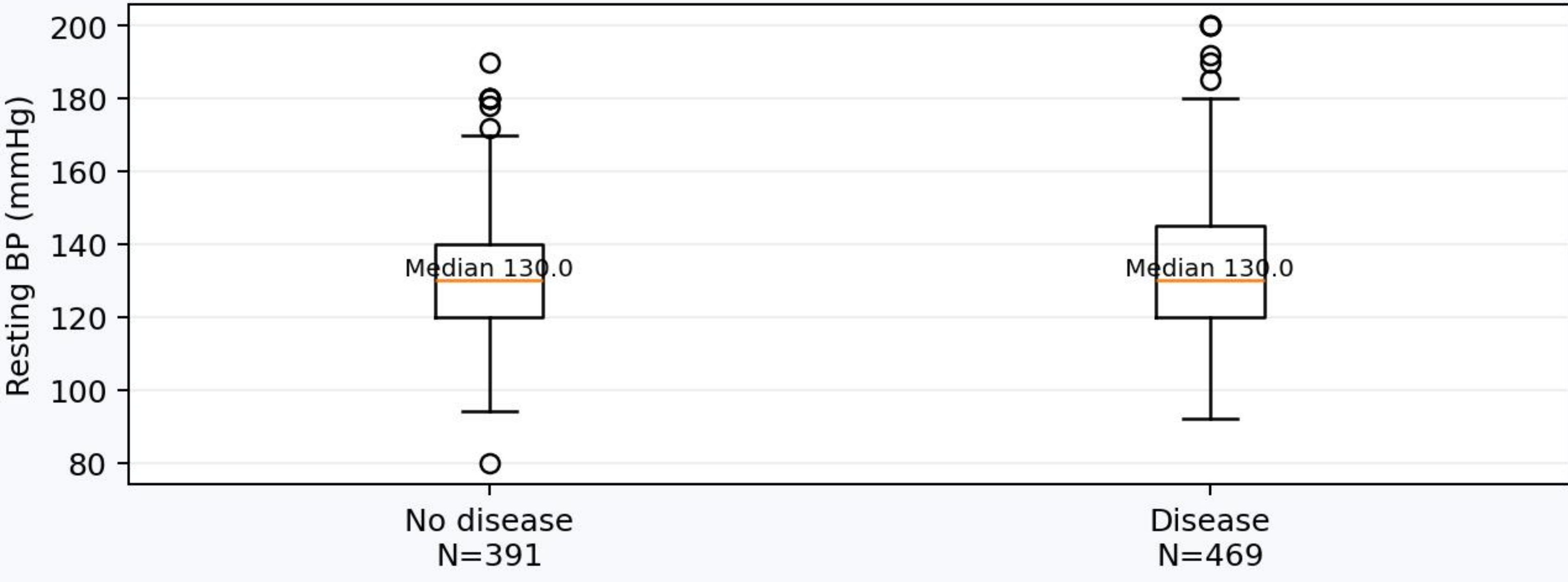
Maximum heart rate is lower in the disease group



Age and maximum heart rate move in opposite directions



Resting blood pressure has substantial overlap



Interpretation: these are associations inside this sample—not causal effects or clinical rules.

Heart Disease Analytics — Data Quality & Model Validation

1,932

Missing values

10 clinical variables

0

Rows removed

conservative cleaning

4

Potential duplicates

flagged, not deleted

0.853

Model sensitivity

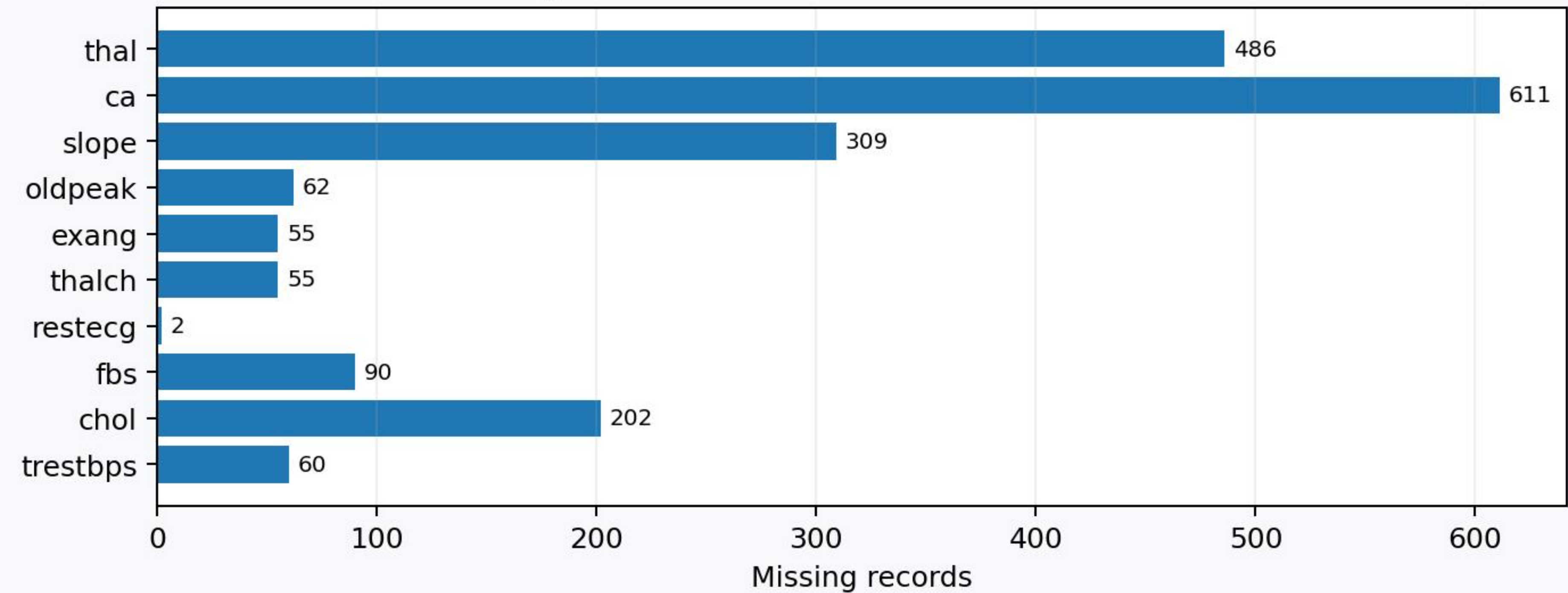
threshold 0.50

0.805

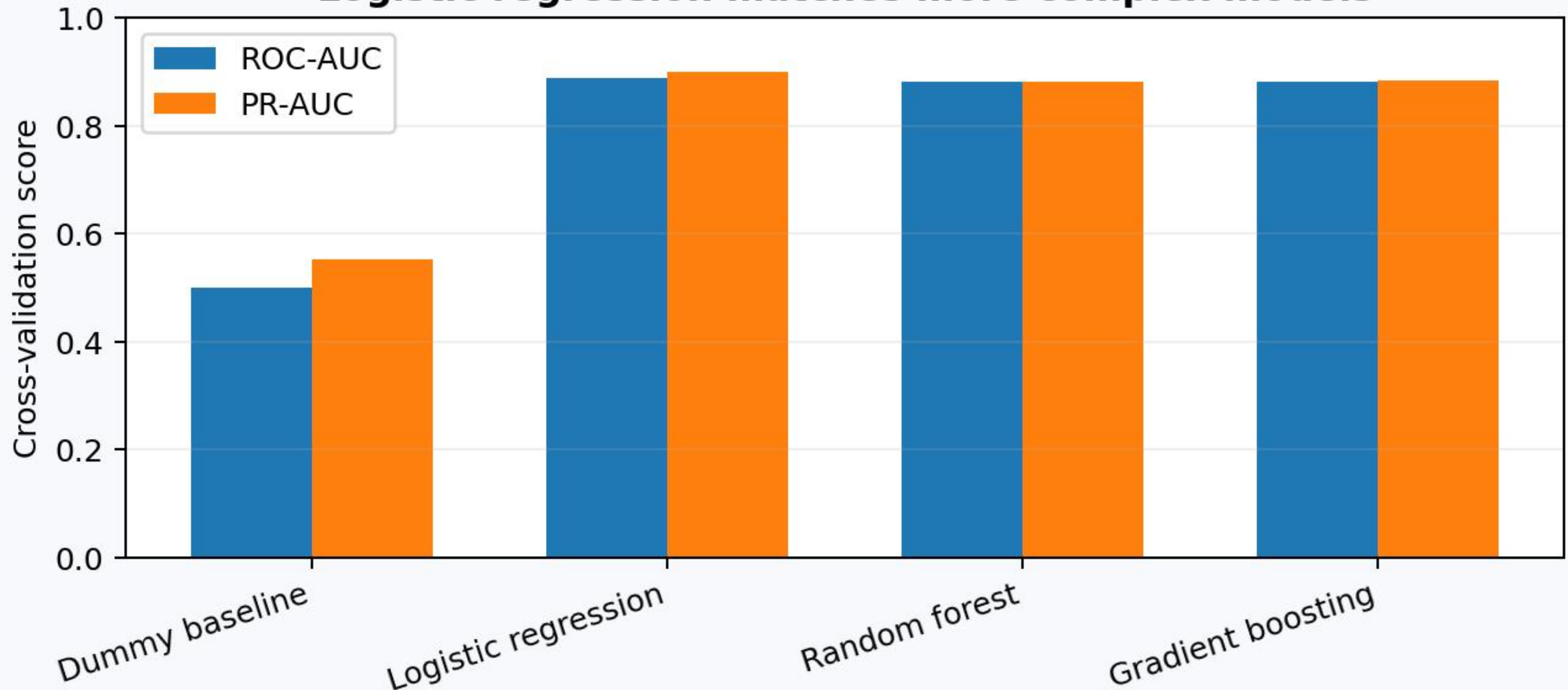
Model specificity

threshold 0.50

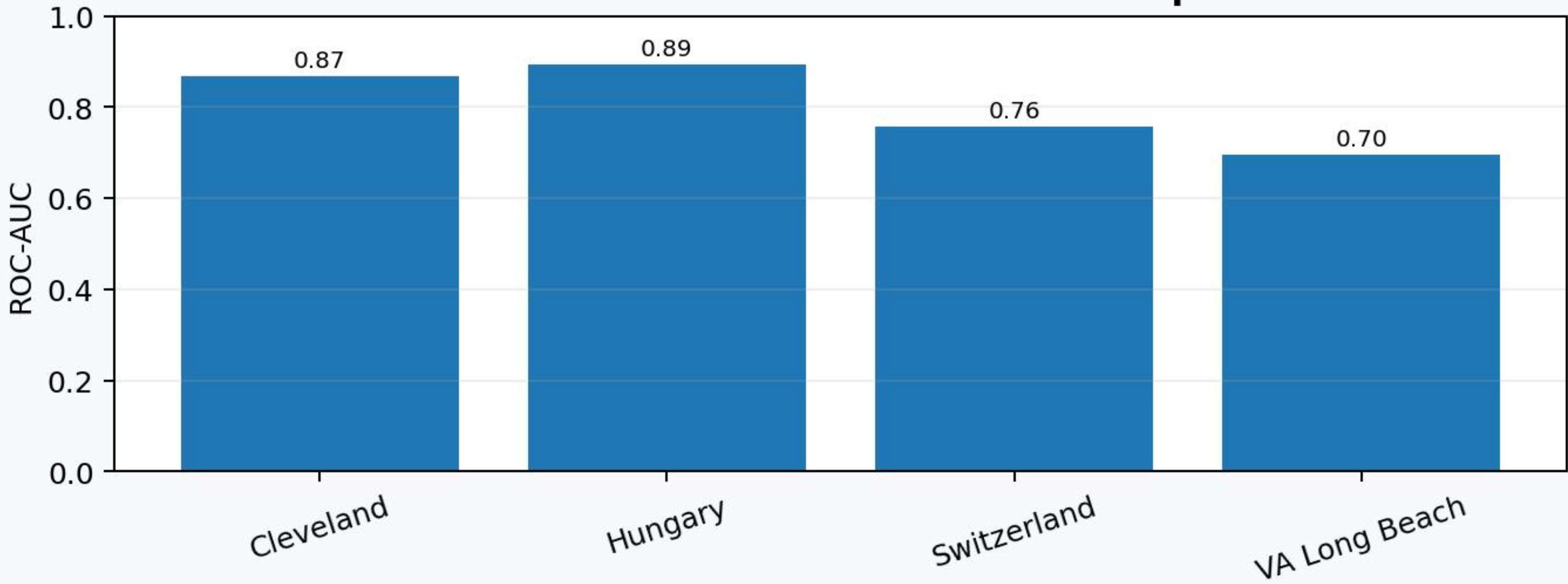
Missingness is concentrated in a few variables



Logistic regression matches more complex models



Generalization weakens on unseen hospitals



Publication decision

- ✓ Publish as a portfolio analytics project.
- ✓ Keep the interpretable logistic baseline.
- ✓ Show source-specific missingness and validation.
- ✗ Do not present the model as clinical decision support.
- ✗ Do not claim that associated features cause disease.

Decision use: judge whether a result is reliable enough to communicate—and whether the model generalizes.